Spatial cell-type annotation verified by KDE: a plain-language tutorial

Oct 29, 2026 · @Razie

1. The big picture

This project gives a cell-type name to every cell in two spatial datasets, then checks each name in three complementary ways: on the tissue, cell by cell with a density model, and against the published experts' labels. The cell-level check is a **consistency diagnostic** that points at cells worth reviewing; it does not prove any label right or wrong.

The problem in one paragraph

Spatial omics measures molecules inside each cell and keeps the cell's position in the tissue. Before any biology can be done, every cell needs a name: tumor cell, T cell, fibroblast, and so on. The workflow used here, and in many labs, groups cells that look alike (clustering) and names each group by its marker genes or proteins. Other approaches exist (reference mapping, per-cell marker scores, probabilistic classifiers), but the cluster-based workflow has a specific weakness: it names **groups, not cells**. Every cell in a group inherits the group's name, including doublets (two cells cut as one), badly segmented cells, and cells caught between two states. The project's question is: **how consistent is each individual cell with the name it inherited?**

The four steps

1. **Annotate the cells myself** with the cluster-based workflow (notebooks 01 and 02).
2. **Check the names on the tissue**: are the named cells arranged the way biology expects (notebooks 01 and 02)?
3. **Check every cell with a density model** (a Gaussian KDE): is this cell consistent with the other cells that carry its name (notebook 03)?
4. **Compare with the original papers**: do my names agree with the experts', and do I reach the same biological findings (notebook 04)?

The two datasets

The same workflow runs on two very different technologies, to show the approach is not tied to one data type.

	Xenium (notebook 01)	CODEX (notebook 02)
What it measures	RNA molecules, counted one by one	proteins, measured as light intensity
Features per cell	313 genes	56 protein markers
Tissue	breast cancer, one section from one patient	colorectal cancer, 35 patients, 140 regions
Cells	167,780	258,385
What a number looks like	a small whole count (often 0)	a brightness value (any positive number)
Original paper	Janesick et al., *Nat Commun* 2023	Schürch et al., *Cell* 2020

Counts and brightness need different handling (different normalization, different QC), which is why each dataset has its own notebook.

The one rule: the answer key stays sealed

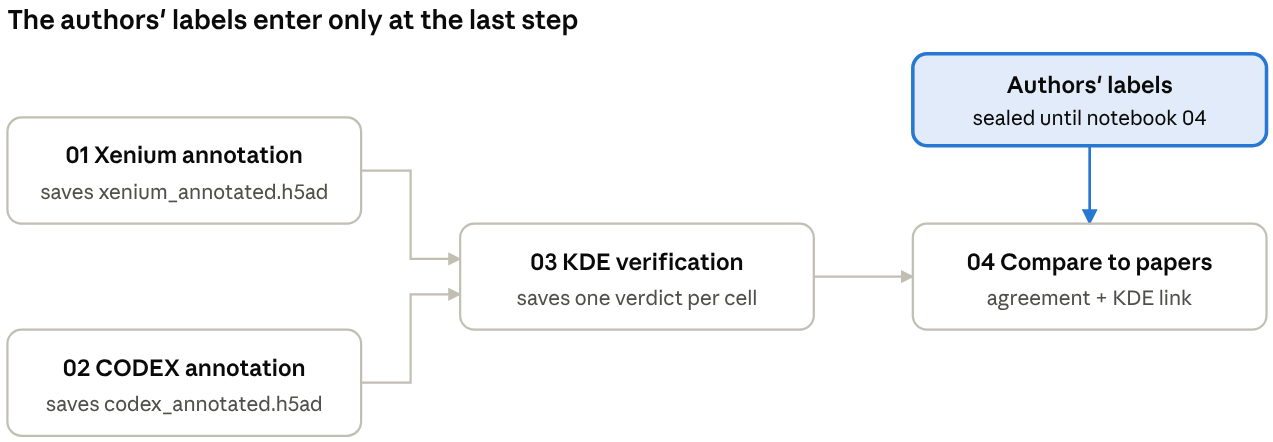
The authors' labels are **never** used to build my annotation or the density model. They are opened only in notebook 04. Think of an exam: if you look at the answer key while answering, agreement with it proves nothing. Keeping it sealed makes the comparison in notebook 04 an honest test.

One exception is recorded openly: after a first look at notebook 04, five CODEX clusters were renamed `unknown` (section 3). The rule for renaming used only notebook-02 evidence, but the CODEX comparison is no longer fully blind, and notebook 04 says so.

How the notebooks pass work along

Each notebook saves its result to a file, so each can be rerun on its own: notebook 01 saves `xenium_annotated.h5ad`, notebook 02 saves `codex_annotated.h5ad`, notebook 03 reads both and saves one verdict per cell (`xenium_kde.csv`, `codex_kde.csv`), and notebook 04

reads everything plus the authors' labels. An `.h5ad` file is an AnnData object on disk: one table of cells × features, plus extra slots for cell information, coordinates and the PCA.



how the notebooks connect · 4 notebooks, 1 sealed input

Notebooks 01 and 02 each feed notebook 03; notebook 04 reads everything before it, and only notebook 04 ever sees the authors' labels.

What the project shows, and what it does not

The five kinds of evidence are complementary; none of them alone establishes a cell's identity.

Evidence	Notebook	What it can show	What it cannot show
Marker coherence	01, 02	each cluster has its own markers	that a cluster is one biological type
Spatial consistency	01, 02	named cells are arranged as expected	identity (biology can break patterns)
KDE consistency	03	which cells fit their label poorly	that a flagged cell is truly mislabeled
Agreement with the papers	04	how often experts gave the same name	ground truth (the reference has its own choices)
Patient-level replication	04 (CODEX)	a finding holds across patients	anything for Xenium (one patient)

The main result, stated at the strength the data support: **KDE-flagged cells were more likely to disagree with the published annotation than confirmed cells, supporting the KDE as a diagnostic for annotation inconsistency.** It does not show that the KDE detects mislabeling accurately.

Reproducibility

Each notebook prints the exact package versions it ran with, and `requirements-lock.txt` pins them (Python 3.13). This matters: run on Python 3.9 with older libraries, the same notebook 01 selects resolution 0.8 (18 clusters instead of 22), and the annotation check stops the notebook, as designed. The hand-written cluster names are valid only for the pinned environment.

2. Notebook 01: Xenium breast cancer

Notebook 01 turns a raw table of RNA counts into 21 named cell types plus `unknown`, then shows that those names make sense on the tissue. Each block below says what the code does, why, and why the numbers are what they are.

Block 1: parameters in one place

Every number the analysis depends on sits in one cell at the top: `MIN_COUNTS = 10`, `N_PCS = 30`, `N_NEIGHBORS = 15`, the resolutions to scan, the coherence thresholds and `SEED = 0`. Anyone can see at a glance what was chosen, and a rerun with the same seed gives the same result.

Block 2: load the data

- **What:** the count matrix (cells × genes) is read from the `.h5` file with `gex_only=True`, which keeps the 313 real genes and drops the control features (negative-control probes and blank codewords). The cell table (`xe1_cells.csv.gz`) adds each cell's centroid position (µm), area and QC counts. Matrix and table are joined on the cell ID, and the code stops if any cell is missing from the table.
- **Result:** one AnnData object: counts in `.X`, cell information in `.obs`, x/y positions in `.obsm["spatial"]`.
- **Why the controls are dropped from the matrix:** they are not biology. They are used once, in the QC block, to measure the false-positive rate.

Block 3: quality control

- **What:** plots of counts per cell, genes per cell and counts vs. cell area. Cells with fewer than 10 gene counts are removed: 164,000 of 167,780 cells stay (3,780 removed).
- **Why 10:** a cell with a handful of transcripts carries too little information to be typed. Ten is the minimum used in the main Xenium benchmark (Salas et al., 2025). A benchmark's choice does not prove the value suits this sample, so the notebook also shows what nearby thresholds would remove: 5 removes 1.7% of cells, 10 removes 2.3%, 20 removes 3.4%, 30 removes 4.9%. Removed cells are smaller than typical (median 90 vs. 140 μm²) but not all tiny, so they mix fragments with cells that captured very little. The threshold is an operational choice.
- **Negative controls:** Xenium includes probes that bind nothing and codewords that should never be read. Any count on them is a false detection. Here they make up 0.09% of all counts, and only 5 cells have more than 10% control counts. The data are clean, so no extra filter is needed.
- **Segmentation caveat:** Xenium draws each cell by growing its nucleus outward (up to 15 μm in the default pipeline). This can hand some transcripts of a neighbor to the wrong cell. It matters most for small immune cells pressed against tumor or stroma, and it is one reason the per-cell check of notebook 03 is useful.

Block 4: normalize and scale

1. **Keep the raw counts** in a layer called `counts`.
2. **Library-size normalization:** each cell is rescaled so its total count equals the median total of all cells. A large cell or a well-captured cell has more counts of everything; this removes that difference.
3. **log1p:** each value becomes log(1 + x). Counts are skewed (a few huge values), and the log compresses them.
4. **Scale:** each gene is set to mean 0 and standard deviation 1, clipped at 10, so a few very high genes do not dominate the next step.

- **Why this recipe:** library size + log + scale was the best-performing preprocessing for clustering Xenium data in a systematic benchmark (Salas et al., 2025).
- **Its known weakness:** with a targeted panel, the total count depends on which genes were put on the panel. A panel rich in one cell type's genes can bias the results (Atta et al., 2024). That is why Block 7 repeats the clustering with a different normalization.
- **No gene selection:** with only 313 hand-picked genes, every gene is informative, so the usual "highly variable genes" step is skipped.

Block 5: PCA, neighbor graph, UMAP

- **PCA** finds the main directions of variation and summarizes 313 genes into a few dozen "principal components". The elbow plot shows how much each one explains; the first 30 explain 35.9% of the variance, and after that each adds little. So `N_PCS = 30`.
- **The kNN graph** links every cell to its 15 most similar cells in that 30-dimensional space. 15 is the scanpy default, a common middle value: fewer makes the graph noisy, more blurs small populations.
- **UMAP** squeezes the graph into 2-D for a picture. It is used only to look; nothing is computed from it.

Block 6: Leiden clustering and the choice of resolution

- **Leiden** cuts the neighbor graph into communities: groups of cells more connected to each other than to the rest. The **resolution** is a dial: higher gives more, smaller clusters.
- **The scan:** Leiden runs at six resolutions (0.2 to 1.2). For each, every cluster gets a **coherence score**: take its 5 top marker genes, and compare the fraction of cells detecting each gene inside vs. outside the cluster.

```
s_k = \frac{1}{5}\sum_{g} \left(\mathrm{frac}_{g,\,\mathrm{in}\,k} - \mathrm{frac}_{g,\,\mathrm{out}\,k}\right)
```

- **In plain words:** a good cluster has genes that are "on" inside it and "off" outside it. If a resolution splits a cell type into pieces with no distinct genes, its weakest cluster's score drops.
- **The rule (coded, not chosen by hand):** take the highest resolution at which every cluster scores at least 0.30. Highest, because a cluster that is split too finely can be given the same name twice, but a cluster that mixes two cell types cannot be undone later.
- **Low-depth clusters:** from resolution 0.6 upward, one cluster of cells with very few transcripts (median about 30) appears. Its "markers" reflect low depth, not identity, so clusters with median counts below 50 are excluded from the rule and named `unknown` later.
- **Result:** the rule selects **1.0**, giving **22 clusters**. A stability check (ARI between neighboring resolutions, 0.80 to 0.90) shows the partition does not jump around between resolutions. Stability is not validity: a coherent cluster can still be a

subdivision of one type, a cell state or a technical effect. The 0.30 threshold is an analyst-defined heuristic, and taking the highest passing resolution leans towards splitting, which naming can undo by giving two clusters one name.

Block 7: does the normalization change the clusters?

- **What:** the whole pipeline is repeated with **cell-area normalization** (counts divided by cell area), the count-free alternative recommended for targeted panels. The two clusterings are compared with ARI (overall agreement, 1 = identical) and, per cluster, the best Jaccard match (overlap divided by union).
- **Result:** ARI 0.72; 14 of 22 clusters match at Jaccard ≥ 0.75, including all lymphoid, DCIS, endothelial and DC clusters.
- **What differs:** area normalization creates one mixed cluster defined by transcript density per μm² rather than by identity, and it moves two boundaries: CAF vs. fibroblast, and proliferating vs. invasive tumor.
- **Decision:** ARI 0.72 is substantial but imperfect agreement, so this is a sensitivity analysis, not proof of invariance. Measured per label (after naming, section 7b of the notebook), 85.9% of cells keep their label. The least stable labels are CAF / myofibroblasts (38%), APOC1+ macrophages (51%) and proliferating invasive tumor (54%), the same borders the KDE flags later. Library size stays, as the benchmark's best workflow, and those names are marked as the least certain.

Block 8: marker evidence

- **Wilcoxon test:** each cluster is compared with all other cells, gene by gene, on log-normalized values. It ranks genes by how consistently higher they are inside the cluster. The top 8 per cluster go into a table, with detection fractions inside and outside.
- **Dot plot of canonical markers:** known genes grouped by expected cell type (EPCAM/KRT8 for epithelium, KRT14/ACTA2 for myoepithelium, PECAM1/VWF for endothelium, CD3E/CD8A for T cells, and so on). Dot size = fraction of cells expressing; color = mean level.
- **Cluster maps:** each cluster drawn alone on the tissue. Position is evidence too: DCIS cells fill round ducts ringed by myoepithelial cells; invasive tumor spreads as a sheet.
- **Caveat:** cells are not independent replicates, so these p-values are rankings, not proof.

Block 9: my annotation

- **What:** a dictionary written by hand: cluster number → (name, key marker), with the evidence in a comment. Example: cluster 6 → ("Invasive tumor", FASN), because of FASN, FOXA1, ABCC11 and its sheet-like position.
- **Unclear clusters:** the low-depth cluster becomes `unknown`.
- **Two safety checks:** every cluster must be named exactly once, and each key marker must be among its cluster's top-10 Wilcoxon genes. Leiden numbers can change with software versions; if they do, the names would land on the wrong clusters, so the notebook stops instead.
- **Check of the naming:** the dot plot is redrawn by cell type; each type should light up its own markers.

Block 10: spatial check

The names came from expression alone. Their position on the tissue is an independent test.

- **Maps:** all types on the whole section, plus a 1.5 mm zoom on the DCIS ducts, centered by the data (median position of DCIS cells), not by hand.
- **Neighborhood enrichment (squidpy):** two cells are neighbors if their centers are within 30 µm (about two cell diameters). For each pair of types, the number of neighbor pairs is compared with 200 random shuffles of the labels. The z-score says how far the real count is above (+) or below (−) chance.
- **Reading the z-scores:** with 160,000 cells, almost every z is huge, so they are used to rank and compare pairs, not as p-values.
- **What it shows:**
  - DCIS cells have myoepithelial cells as their top neighbors (z ≈ 130–140), while invasive tumor avoids them (z ≈ −97). Loss of the myoepithelial layer is exactly what separates invasive from in-situ cancer, and it was recovered without using position to name cells.
  - Endothelial and perivascular cells pair up (vessels).
  - CD4+ T and B cells cluster together (lymphoid aggregates).
  - `unknown` is compact at the tissue edge, which fits an edge artifact.

All of this is descriptive and comes from one section of one patient. It shows the labels are spatially consistent, not that they are correct, and nothing here is replicated across patients.

Block 11: save

`results/xenium_annotated.h5ad` stores the names, clusters, PCA and positions for notebooks 03 and 04.

3. Notebook 02: CODEX colorectal cancer

Notebook 02 runs the same workflow on protein data and ends with 14 named cell types plus `unknown` (about a third of the cells). It also shows that a common spatial test can mislead when the data come from many patients. The big differences from notebook 01: numbers are brightness, not counts; every region was imaged separately; and 35 patients sit in one table.

Block 1: parameters

The key numbers are `COFACTOR = 150`, the QC thresholds (`BRIGHT_PCT = 95`, `MAX_BRIGHT = 20`, `NUC_PCT = 1`), the background percentile `BG_PCT = 25`, `N_PCS = 20`, `N_TOP = 3`, `LOW_SIGNAL = 0.6` and `MIN_COHER = 0.75`. The resolution is left to the coherence rule in code. The list `LINEAGE` holds the 34 markers that define cell identity (CD3, CD20, Cytokeratin, aSMA and so on).

Block 2: load and check the data

- **What:** one CSV, one row per cell. Marker columns are recognized by `:Cyc_` in their name. The two nuclear stains (HOECHST, DRAQ5) are set aside for QC only. Coordinates `X:X`, `Y:Y` are in pixels and **per region**: every region starts at 0, so cells from different regions can never be compared by position.
- **Metadata kept:** patient, group (1 = CLR, 2 = DII), region, cell size. The authors' cell-type and neighborhood columns are deliberately not read.
- **Checks:** the code stops if a required column is missing, a lineage marker is absent, marker names or cell IDs repeat, a group code is unexpected, or any value is not a finite number. Better to stop loudly than to analyze broken input quietly.
- **CLR vs. DII:** CLR (Crohn's-like reaction) tumors are surrounded by lymphoid aggregates; DII (diffuse inflammatory infiltration) tumors have immune cells spread around. 17 CLR and 18 DII patients.

Block 3: quality control (debris)

The authors' segmentation already removed small objects, so the size and nuclear-stain plots show no separate debris population. Two simple rules remove what is left:

- **Weak nuclear stain:** a real cell has a nucleus. Cells below the 1st percentile of HOECHST are dropped (2,584 cells).
- **Pan-bright objects:** a real cell is bright in a few lineage markers. Autofluorescent debris glows in many unrelated channels at once. A cell above its 95th percentile in 20 or more of the 56 markers is dropped (1,172 cells).
- **Result:** 254,639 of 258,385 cells stay. Anything left with no clear identity becomes its own cluster and is named `unknown` later.
- **Why so gentle:** stricter thresholds would remove real but rare cells. Notebook 04 checks afterwards that removed cells are enriched for what the authors call "dirt".

Block 4: normalize

1. **arcsinh(x / 150):** behaves like a straight line near zero and like a logarithm for bright values. It plays the role log1p plays for counts. The cofactor 150 is the usual value for fluorescence intensities; it sets where "dim" ends and "bright" begins.
2. **Per-region background correction:** each region was stained and imaged separately, so a whole region can come out brighter or dimmer in every channel. For each region and marker, the 25th percentile is subtracted and negatives are set to 0. A low percentile is used because for nearly every marker most cells in a region are negative, so that value is background, not signal.
3. **Scale** each marker to mean 0, standard deviation 1 (clipped at 10).

- **Does it work?** The code measures how much of each principal component is explained by the region (R^2). For PC1 it falls from 0.29 to 0.09: most of the region-to-region shift is gone.
- **Honest caveat:** this correction can also remove real biology. For example, it could remove signal from a region that is truly rich in one cell type.
- **Sensitivity test (section 7c of the notebook):** the whole pipeline is repeated with no correction, the 10th and the 50th percentile. The clustering changes substantially (ARI 0.50–0.55 with the main one), and only 70–72% of named cells keep their label. The CODEX annotation therefore depends on this preprocessing choice much more than the Xenium annotation depends on its normalization (86% stable). Conclusions that rest on specific CODEX labels inherit this uncertainty.

Block 5: PCA on lineage markers only

- **What:** PCA uses only the 34 lineage markers (`mask_var="lineage"`). The 22 others (Ki67, PD-1, ICOS, EGFR, CD5…) describe cell state or have high diffuse background; they stay in the data for description.
- **Why:** in a first test with all markers, high-background channels dominated the clusters and the region effect was strong. Clustering on lineage markers is also what the original authors did.
- **N_PCS = 20:** the elbow flattens there, and the first 20 PCs hold 85.8% of the lineage-marker variance (34 markers need fewer PCs than 313 genes). Then the kNN graph (15 neighbors) and UMAP, as in notebook 01.

Block 6: Leiden and the choice of resolution

- **Coherence for intensities:** "fraction of cells detecting a gene" has no meaning for brightness, so it becomes the **standardized mean difference**: for the cluster's 3 top lineage markers, the mean scaled value inside minus outside. The unit is the standard deviation, and the threshold is 0.75 SD (a clear, visible separation).
- **Low-signal clusters:** a cluster whose cells are dim in everything (median total signal below 0.6 × the overall median) has no real markers and is excluded from the rule, like the low-depth cluster in notebook 01.
- **The rule:** highest resolution where all normal-signal clusters reach 0.75. Only 0.4 and 1.0 pass, so the rule selects **1.0**, giving **24 clusters**.
- **The weak spot:** at 0.8 and 1.2, the failing cluster is always the same dim group with weak CDX2 and β-catenin. Its signal sits right at the low-signal cut, so the resolution choice is sensitive to `LOW_SIGNAL`. The notebook says so openly.
- **How fragile is the choice?** `LOW_SIGNAL` and `MIN_COHER` are operational values, not biological cutoffs. Re-applying the rule on a grid of nearby values moves the selected resolution between 0.4 and 1.5, and the strictest settings select none. Resolution 1.0 is chosen only at the default pair, so the clustering granularity, and with it the annotation dictionary, is tied to one point of that grid.

Block 7: marker evidence

A heatmap of the mean lineage-marker intensity per cluster (scaled 0–1 per marker), a table of mean z-scores, a separate heatmap of functional markers, and the UMAP. The table matters when two clusters look alike in the heatmap.

Block 8: my annotation (the conservative rule)

- **The rule:** a cluster with low signal, weak coherence or **conflicting lineage markers** is called `unknown` rather than forced into a type.
- **Five clusters became `unknown`** under this rule:
  - the dim CDX2+ group;
  - a CD21-high group without CD20, so not clearly follicular DCs;
  - a CD38-high group and a CD138-high group, both with epithelial CDX2/β-catenin signal, so it is unclear whether they are plasma cells or epithelium;
  - a Granzyme B-high group with CD68 but no T or NK markers.
- **Kept with a caution:** Nerves (provisional) and Tregs (could be activated CD4+ T cells).
- **Cost:** `unknown` holds 84,691 cells (33%). That is an honest price: a wrong name spreads errors into every later analysis, while `unknown` simply drops out.
- **Safety checks:** each cluster named exactly once; each key marker in its cluster's top 5 lineage markers; every name has a color. Any failure stops the notebook.

Where the `unknown` cells come from. A high `unknown` share can mean a careful annotation, but also insufficient marker coverage, preprocessing problems or an overly strict rule, so it is not evidence of quality by itself. The breakdown in the notebook shows:

- **A cluster-level decision:** every `unknown` cell comes from a cluster named `unknown` as a whole.
- **By signal:** `unknown` is much more common among the dimmest fifth of cells (47%, against about 30% in the other quintiles).
- **By patient and region:** the share varies from 16% to 60% per patient, but is similar in CLR and DII (median 35% vs. 36%). Patient-level comparisons therefore rest on different fractions of each patient's cells.

Block 9: spatial check, done within regions

- **Graph:** built separately in each of the 140 regions (`library_key="region"`), radius 30 µm (pixel size 0.377 µm), so cells from different regions are never neighbors.
- **The correction that matters:** squidpy's standard test shuffles labels across **all** cells. With 140 regions of very different make-up, that mixes two things: cells being next to each other, and cell types simply living in the same patients' regions. Here labels are shuffled **within each region** (200 times), keeping each region's composition fixed, so the z-score measures only the local arrangement.
- **How big the difference is:**

Pair	Shuffled across all regions (z)	Shuffled within regions (z)
B cells next to CD4+ T cells	140	11
B cells next to B cells	911	183
Nerves next to Nerves	544	7

- **Reading:** B-cell aggregates and tumor structure are real local patterns. The B–T pairing was mostly "same patients", and "Nerves" have almost no spatial structure, which weakens that label further.
- **Limit:** these are cell-level descriptions, not patient-level tests, and closeness is association, not interaction.

Significance. For every pair of types the notebook now also computes an empirical two-sided permutation p-value (the share of shuffles at least as extreme as the real count) and corrects it for testing all 120 pairs (Benjamini–Hochberg). With 200 shuffles the smallest possible p is about 0.01, and 109 of 120 pairs reach $q < 0.05$. At the cell level almost every pair departs from random, so significance says little; the z-scores remain the useful ranking, and inference about spatial organization would need a patient-level test.

Block 10: save

`results/codex_annotated.h5ad` stores names, clusters, PCA, coordinates and patient information.

4. Notebook 03: checking every cell with a KDE

Notebook 03 moves the check from the cluster to the single cell. It is a **consistency diagnostic**: it finds 94.6% of Xenium cells and 91.5% of CODEX cells consistent with their label, and flags the rest for review. A flag means "fits its label poorly in this space", not "wrong". The authors' labels are still not used.

The idea in plain words

Take all cells I called "B cells". Put a small, smooth bump (a Gaussian) on each one in PCA space and add the bumps up. You get a landscape: high where B cells crowd together, low where they are rare. That landscape is a **kernel density estimate (KDE)** of $p(x \mid \text{B cell})$. Build one landscape per cell type. Now take any cell:

- **Its height on its own type's landscape** says how typical it is of its own type.
- **Its heights on the other landscapes** say whether another type explains it better.

Each cell then gets one of three verdicts:

Verdict	Rule in words
fits another type	another type is at least 10 times more probable, **and** the cell looks typical for that other type
outlier	the cell sits lower on its own landscape than 99% of typical cells of its type
confirmed	neither

"More probable" uses Bayes' rule: the posterior p(c | x) combines each landscape's height with the type's frequency (the prior). Everything is computed in logarithms (`logsumexp`) because the raw densities are tiny numbers that would round to zero.

Block 1: parameters and why

- `K = 10` **PCs:** a KDE gets unreliable as dimensions grow (points spread out and every cell looks lonely). Ten PCs keep the main structure and stay workable. 5 and 20 are tested later.
- `N = 1000` **support cells per type:** a random sample of each type builds its landscape, so the run stays laptop-fast. 500 and 2000 are tested later.
- `N_VAL = 500` **held-out cells per type:** used to pick the bandwidth and the outlier threshold, never to build the landscape.
- `Q = 1%` **outlier threshold:** the 1st percentile of held-out cells' own-type height. A strict cut: only clearly atypical cells.
- `ODDS = 10` **mislabel threshold:** another type must be 10 times more probable. This is a common convention for "strong" evidence; it ignores mild doubt.
- `TILE_UM = 500` **Xenium fold tiles:** 500 µm is several tissue structures wide, large enough that neighboring cells rarely sit in different folds.

Block 2: whitening (shaping the bumps)

- **The problem:** PC1 varies far more than PC10, and each cell type is stretched in its own directions. scikit-learn's `KernelDensity` uses round bumps of one size in all directions.

- **The fix:** before building a type's landscape, its cells are **whitened**: centered on the type mean and rescaled by the type's own covariance, so the cloud becomes round. A round bump in the whitened space equals a bump with the type's own shape in the original space (bandwidth matrix $H = h^2 \Sigma$).

```
y = L_c^{-1}(x - \mu_c), \qquad \hat\Sigma_c = L_c L_c^{\top}, \qquad \log
p(x \mid c) = \log p_y(y \mid c) - \log\lvert\det L_c\rvert
```

- **Why the last term matters:** rescaling stretches or squeezes space, which changes heights. Subtracting log|det L| (the Jacobian) puts every type's landscape back on the same scale, so types can be compared fairly.
- **No leakage:** the mean and covariance are estimated from the support cells only, never from the held-out cells.

Block 3: bandwidth (how wide the bumps are)

- **Why it matters most:** too narrow and the landscape is a field of spikes around the support cells, so every new cell looks atypical. Too wide and all structure blurs.
- **Start:** Scott's rule, $h_0 = n^{(-1/(K+4))}$, the textbook width for round data.
- **Choice:** five candidates (0.5, 0.75, 1, 1.5, 2 × h_0) are tried; the one under which the held-out cells get the highest average log-likelihood wins. The data pick the width.
- **Result:** 1 × Scott in almost every fit, so the textbook width was already right here.

Block 4: cross-fitting (no cell grades itself)

- **The problem:** a cell that helped build its own landscape always sits on a peak, so it would always look typical.
- **The fix:** split the cells into two halves; build landscapes on one half, score the other; then swap.
- **How the halves are cut:** Xenium by 500 µm tiles (176 tiles assigned at random); CODEX by patient, half the patients of each group per fold, all regions of a patient together. Neighbors and same-patient cells are often near-copies, so splitting this way stops them vouching for each other.

What cross-fitting does not cover. It applies to the KDE step only. The PCA and the cluster labels were computed on the full dataset, so held-out cells did influence the space and the clusters their labels come from. The check therefore asks whether a cell fits an already-established label; it is not an end-to-end validation of the annotation pipeline.

That would mean fitting PCA and clustering on training data and assigning held-out data by a fixed rule, which is out of scope. Tiles also leave cells near a tile edge with close neighbors in the other fold; the notebook tests this (below).

Block 5: the toy example

Two made-up classes in 2-D: A stretched, B round. The panels show A's landscape at three bandwidths (spiky, chosen, blurred), the line where A and B are equally likely, the lines where one is 10 times more likely, and the outlier zone. Three test points labeled "A" get the three verdicts.

The point far from both classes shows why "fits another type" also requires the competitor to find the cell typical. Far from all data, the ratio between two landscapes is decided by whose tail falls off more slowly, which says nothing about the cell. Such a cell is an outlier, not a member of the other class.

Block 6: results on the real data

- **Overall:** Xenium 94.6% confirmed, 0.4% outliers, 5.0% fit another type. CODEX 91.5%, 1.2%, 7.3%.
- **Xenium:** rare types are often better explained by a common neighbor type: mast cells 34% (toward fibroblasts), adipocytes 27%, mature DCs 24% (toward CD4+ T), NK cells 22% (toward CD8+ T), pDCs 17%.
  - It is not the prior punishing rare types: with equal priors, 97–98% of verdicts stay the same.
  - More likely, their identity rests on a few genes that carry little weight in the first 10 PCs.
- **Also flagged in Xenium:** borders between related states: DCIS 1 → invasive tumor, proliferating → invasive, CAF ↔ fibroblasts. The profile heatmaps show flagged cells sitting between the two types, as transitions would.
- **CODEX questioned labels:** 46% of Nerves fit another type better (toward CD8+ T cells) and Tregs 12% (toward CD4+ T), against a median of 6.5% for other types.
- **Outputs:** flag rates per type, flag maps on the tissue, and profile heatmaps comparing flagged cells with confirmed cells of both types.

Robustness across random folds and samples. The whole cross-fitting was repeated 5 times, each with a new fold assignment and new support samples, scoring the same 600 cells per type:

- **Overall:** 86% (Xenium) and 79% (CODEX) of these cells got the same verdict in every repeat.
- **Stable types:** for most types the flag rate moves by only a few percentage points.
- **Unstable types:** adipocytes (24–41%), Nerves (31–42%) and MUC-1+ tumor cells (12–21%). For Nerves only 42% of cells keep the same verdict, so single-run flags for these types should not be over-read.

Tile edges. Cells within 30 µm of a tile edge are flagged 5.6% of the time, against 5.4% farther inside. Near-copies in the other fold do not make edge cells look more typical, so the tile blocking is adequate here.

Block 7: sensitivity

Each variant fits on fold 0 and scores the same 20,000 cells from fold 1, changing one setting at a time: K (5, 20), fixed bandwidths (0.5, 1, 2 × Scott), N (500, 2000), Q (0.5%, 5%) and uniform priors. Across all of them, 82–100% of verdicts match the baseline. The only setting that shakes them is a too-narrow bandwidth (0.5 × Scott), which the held-out choice avoids.

Limits stated in the notebook

- **Built-in circularity:** the names come from clusters built in the same PC space, so most cells agree with their type almost by construction. The check is informative mainly at cluster borders and for small types. It cannot catch a wrong name shared by a whole cluster.
- **Consistency, not truth:** a flag means "this cell does not fit its label well in this space". It does not prove a doublet or a wrong call.
- **Model-based numbers:** the posterior uses label frequencies as priors and the model's own density estimates. It is not a calibrated biological probability, and the 1% and 10:1 thresholds are operational definitions, not error rates.
- **Rare types:** a high flag rate for a small type can come from a heterogeneous class, key markers that weigh little in the first 10 PCs, too few support cells, overlap with a related type, or a wrong label. The KDE cannot tell these apart; the marker heatmaps are the evidence.

5. Notebook 04: comparing with the original papers

Notebook 04 opens the answer key. My Xenium labels agree with the authors' for 75% of cells and reproduce the paper's tumor structure. My CODEX labels reproduce the central finding (B-cell follicles in CLR patients), and cells flagged by the KDE disagree with the experts far more often than confirmed cells.

Block 1: the name mapping, written before looking

- **The problem:** the two annotations use different names and different detail. The authors have one broad "Stromal" class; I split fibroblasts, CAFs and adipocytes. They split macrophages into six classes; I have one or two.
- **The fix:** both sides are mapped onto a **common vocabulary**, written from the names and marker evidence alone, before any agreement is computed. Mapping after seeing the agreement would let me tune the map until it looks good.
- **Examples:**
  - Both DCIS groups map to one `DCIS` class, because the numbers 1 and 2 are arbitrary on both sides.
  - The authors' LAMP3+ DCs map to my mature DCs, and their IRF7+ DCs to my pDCs (IRF7 is the master switch of pDCs).
  - The authors' "Unlabeled", "dirt", "undefined" and mixed classes, and my `unknown`, all map to `unassigned`.
- **Checks:** the code stops on any author label or my label that the map does not cover, and on duplicate cell IDs.

Block 2: cell-level agreement

- **Coverage first:** agreement is computed only on cells that both sides name (neither is `unassigned`). The coverage line reports how many that is, because moving doubtful cells to `unknown` removes them from the count and can make agreement look better.
- **ARI (adjusted Rand index):** asks, for every pair of cells, whether both annotations put them together or apart. It runs from 0 (chance) to 1 (identical) and does not depend on the names.
- **Per-type agreement in both directions:** "% of mine they agree with" and "% of theirs I recover". A broad class scores high one way and low the other, which is how granularity differences show up.
- **Confusion matrix:** for each of my types, where the authors put those cells.

	Xenium	CODEX
Cells named by both sides	94.2%	64.0%
ARI (common vocabulary)	0.55	0.37
Same name	75.3%	59.3%

- **Xenium:** most disagreement is granularity. The authors' "Stromal" holds 27–37% of my immune cells. I recover 92–100% of their CD4+ T, B, LAMP3+ DC, IRF7+ DC and mast cells.
- **CODEX:** agreement is lower, for two systematic reasons. First, the stroma/smooth-muscle border falls in a different place. Second, the authors call about twice as many cells macrophages.

Keeping `unassigned` as a category. The main metrics exclude cells either side left unnamed (9,507 Xenium cells, 91,548 CODEX cells). Counting `unassigned` as a label of its own gives Xenium ARI 0.52 and 71.3% same label, and CODEX ARI 0.17 and 43.2%.

Primary (blind) vs. post hoc CODEX labels. The first version of notebook 02 named the five later-revised clusters, before any comparison with the authors, so it is the **primary, blind** annotation. The revised labels are a **post hoc** analysis.

CODEX	Primary: first-version labels	Post hoc: revised labels
Cells named by both sides	78.4%	64.0%
ARI, named cells	0.371	0.372
Same label, named cells	57.3%	59.3%
ARI, `unassigned` kept	0.246	0.172
Same label, all cells	47.4%	43.2%

The revision did not make the annotation agree better with the authors. It mainly removed cells from the comparison. The primary result is the first column.

Block 3: composition and QC

- **Composition:** bar charts of my vs. the authors' cell-type proportions on the common vocabulary.

- **Xenium QC:** all 3,780 cells my QC removed are cells the authors also left "Unlabeled". The two pipelines agree on what is too poor to type.
- **CODEX QC:** the cells I removed are 7.3% "dirt" by the authors' labels, against 2.8% overall (2.6 times enriched). But they catch only 275 of the authors' 7,357 dirt cells; most others ended up in my `unknown`.
- **The questioned clusters:** a table shows what the authors call each cluster that notebook 02 renamed `unknown`.
  - CD21-high: 49% macrophages, so `unknown` was right.
  - CD138-high: 63% tumor, so `unknown` was right.
  - CD38-high: 42% plasma cells, and dim CDX2+: 61% tumor. For these two the old names were partly right, and the conservative rule cost some correct labels.

Block 4: the papers' main findings

Xenium (Janesick et al.):

- **DCIS vs. invasive:** among cells both sides call tumor, 96.4% fall in the same compartment, and the maps of both annotations are nearly identical.
- **The two DCIS subgroups:** found independently, with swapped numbers. My DCIS 1 is their DCIS_2 (98% of cells), my DCIS 2 their DCIS_1 (93%).
- **Immune populations:** all those the paper reports are recovered, including the rare LAMP3+ DCs (99%) and IRF7+ DCs (94%).

CODEX (Schürch et al.), with the patient as the unit:

- **Group test:** for each patient, the fraction of each immune type; then CLR vs. DII with a Mann–Whitney test (rank-based, no assumption of normal data). The test is done twice, with my labels and with theirs.
- **Multiple-testing correction:** seven types are tested, so p-values get a Benjamini–Hochberg correction into q-values. Without it, about one test in twenty looks significant by luck.
- **Result:** B cells are enriched in CLR with both labelings ($q = 0.007$ mine, 0.005 theirs). No other type passes $q < 0.05$ with either labeling. Before correction, macrophages were significant with the authors' labels; after it they are not.
- **Follicles:** I call a B cell "in an aggregate" when at least half its neighbors within 30 µm are B cells.
  - These aggregate cells make up a median 4.2% of cells in CLR patients and 0.0% in DII ($p = 0.0004$).

  - The authors' "Follicle" neighborhood gives 3.8% vs. 0.02% ($p = 0.0003$).
  - 82% of my aggregate B cells lie inside their Follicle neighborhood, against 5% of all cells.
- **Caveat:** my rule is not the authors' neighborhood algorithm, so this shows similarity, not equivalence.

Block 5: does the KDE do real work?

The key test of the project. If the KDE flags are meaningful, flagged cells should disagree with the experts more often than confirmed cells.

	Xenium	CODEX
Confirmed cells disagreeing with the authors	24%	39%
Cells that fit another type, disagreeing	42%	63%
Odds ratio, crude	2.27	2.36
Odds ratio, stratified by cell type	2.91	2.84
KDE's suggested type = authors' label	60%	28%

- **Why stratify (Mantel–Haenszel):** some types get flagged a lot and also disagree a lot for unrelated reasons, such as granularity. A crude odds ratio could then come from type mix alone (Simpson's paradox). The Mantel–Haenszel ratio compares flagged vs. confirmed **within** each type and pools the results. It got stronger, not weaker.
- **Why a patient-level test:** the Fisher p-value (about 10^{-259}) treats 150,000 neighboring cells as independent, which they are not. So for CODEX each patient is compared on its own. In all 35 of 35 patients, flagged cells disagree more, by a median of 21.6 percentage points (Wilcoxon signed-rank $p = 5.8 \times 10^{-11}$).
- **Reading:** KDE-flagged cells were more likely to disagree with the reference annotation than confirmed cells, supporting the KDE as a diagnostic for annotation inconsistency. The flags were computed before the reference labels were opened (in notebook 04, the CODEX flags use the revised labels).

Limits stated in the notebook

- **Granularity:** confirmed cells still disagree 24% (Xenium) and 39% (CODEX) of the time. Mostly this is a difference in detail, which the KDE cannot see because it checks cells only against my own types.
- **Not ground truth:** the authors' labels are a reference made by other people with other choices, so disagreement does not prove I am wrong.
- **Not fully blind:** the CODEX relabeling happened after a first look at this comparison.
- **Association, not accuracy:** the odds ratios measure an association between flags and reference disagreement. They are not the sensitivity, specificity or positive predictive value of the KDE for truly wrong labels, because no ground truth exists.
- **Conditional comparison:** the KDE link uses only cells both annotations name.
- **Findings vs. labels:** reproducing the papers' main findings supports consistency with the published biology, not the correctness of every cell label.
- **Xenium is one patient:** there is no patient-level replication for the Xenium results.

6. Cheat sheet: every number and why

Notebook	Parameter	Value	Why this value
01	`MIN_COUNTS`	10 transcripts	fewer is too little to type; the Xenium benchmark uses 10
01	normalization	median library size, log1p, scale (clip 10)	best clustering workflow in the Xenium benchmark; area normalization tested as a check
01	`N_PCS`	30	elbow of the PCA curve; 35.9% of variance
01, 02	`N_NEIGHBORS`	15	scanpy default; balances noise and small populations
01	`N_TOP`	5 genes	enough to describe a cluster without diluting its best markers
01	`MIN_DEPTH`	50 counts (median)	below it, a cluster's "markers" reflect low depth, not identity

Notebook	Parameter	Value	Why this value
01	`MIN_COHER`	0.30	every cluster keeps genes clearly more often on inside than outside
01, 02	resolution	1.0 (selected by the rule)	highest resolution where every normal cluster stays coherent
01, 02	spatial radius	30 µm	about two cell diameters: direct neighbors
01, 02	`N_PERMS`	200	enough shuffles for a stable z-score at laptop speed
02	`COFACTOR`	150	the usual arcsinh cofactor for fluorescence intensities
02	`NUC_PCT`	1st percentile of HOECHST	no nucleus, no cell; gentle so rare real cells stay
02	`BRIGHT_PCT`, `MAX_BRIGHT`	95th percentile, 20 of 56 markers	real cells are bright in few markers; debris glows everywhere
02	`BG_PCT`	25th percentile per region	most cells are negative for any marker, so this value is background
02	lineage markers	34 of 56	identity markers only; state and high-background channels left out
02	`N_PCS`	20	elbow; 85.8% of lineage-marker variance
02	`N_TOP`	3 markers	protein panels have few markers per type
02	`MIN_COHER`	0.75 SD	a clear, visible separation in standardized units
02	`LOW_SIGNAL`	0.6 × median total signal	dim-everywhere clusters have no real markers

Notebook	Parameter	Value	Why this value
03	`K`	10 PCs	a KDE weakens as dimensions grow; 5 and 20 tested
03	`N`	1,000 cells per type	fast and stable; 500 and 2,000 tested
03	`N_VAL`	500 cells per type	held out for bandwidth and threshold, never for fitting
03	`H_GRID`	0.5–2 × Scott	the data choose the width; 1 × Scott won almost always
03	`Q`	1%	only clearly atypical cells become outliers
03	`ODDS`	10	10:1 is the usual threshold for strong evidence
03	`TILE_UM`	500 µm	wide enough that neighbors share a fold
03	`SENS_N`	20,000 cells	same cells for every variant; fast
04	`AGG_MIN`	half of neighbors are B cells	a simple, readable definition of a B-cell aggregate
04	significance	BH q < 0.05	corrects for testing several cell types at once
all	`SEED`	0	the same numbers on every rerun

Every threshold above is an operational choice, not a biological constant. The ones that matter most are tested in the notebooks: the QC threshold (nearby values), normalization (area vs. library size), the coherence thresholds (a grid), the background correction (three alternatives) and every KDE parameter.

Added in the review round

Notebook	Parameter	Value	Why this value
01	`MIN_COUNTS` alternatives	5, 10, 20, 30	shows how many cells each threshold would remove
02	`LOW_SIGNAL` × `MIN_COHER` grid	0.4–0.8 × 0.5–1.0	shows how the selected resolution depends on the thresholds
02	background alternatives	none, 10th, 50th percentile	shows how much the labels depend on the correction
02	permutation p-values	200 shuffles, BH over 120 pairs	formal test; smallest attainable p ≈ 0.01
03	`N_SEEDS`	5 repeats	new folds and support samples each time
03	`EVAL_PER_TYPE`	600 cells	the same cells scored in every repeat; fast
03	`EDGE_UM`	30 µm	one neighbor radius from a tile edge
04	patient minimum	20 eligible cells per group	patients with too few flagged or confirmed cells are left out and counted
all	pinned versions	`requirements-lock.txt`	Leiden results depend on library versions

Glossary

- **AnnData / .h5ad:** the standard Python container for single-cell data: a cells × features table plus cell information, coordinates and results.
- **QC (quality control):** removing cells too poor to analyze.
- **Normalization:** making cells comparable when some captured more molecules or are larger.
- **PCA:** summarizes many correlated features into a few directions of variation.
- **kNN graph:** each cell linked to its k most similar cells.
- **Leiden:** splits that graph into communities (clusters); the resolution sets how many.

- **UMAP:** a 2-D picture of the graph; for looking only.
- **Marker:** a gene or protein typical of one cell type.
- **Wilcoxon / Mann–Whitney test:** compares two groups by ranks, without assuming normal data.
- **Neighborhood enrichment:** are two types neighbors more often than random shuffles would give?
- **z-score:** how many standard deviations a value lies from its expected value.
- **KDE:** a smooth density built from many small bumps, one per cell.
- **Bandwidth:** the width of those bumps.
- **Whitening:** rescaling data so its spread is round; makes one bandwidth fit all directions.
- **Posterior / prior:** the probability of a type given the cell, and the type's overall frequency.
- **Cross-fitting:** fitting on one half, scoring the other, so no cell grades itself.
- **ARI:** agreement of two clusterings, from 0 (chance) to 1 (identical).
- **Jaccard index:** overlap of two groups divided by their union.
- **Odds ratio:** how much more likely an outcome is in one group than another (1 = no difference).
- **Mantel–Haenszel:** an odds ratio computed within strata (here cell types) and pooled, removing confounding by type.
- **Benjamini–Hochberg (BH):** corrects p-values when many tests run at once; gives q-values.
- **Pseudoreplication:** treating correlated cells as independent samples, which makes p-values far too small; avoided by testing per patient.
- **Fits another type:** the KDE verdict for a cell that another type explains at least 10 times better, and that is typical for that type. Formerly called "possibly mislabeled"; renamed because a flag is not a verified error.
- **Primary vs. post hoc:** the primary analysis uses choices fixed before the reference was seen; a post hoc analysis uses choices made after.

References

- Janesick A. et al. *Nat Commun* 14, 8353 (2023), the Xenium dataset and labels.
- Schürch C. M. et al. *Cell* 182, 1341–1359 (2020), the CODEX dataset and labels.
- Salas S. M. et al. *Nat Methods* (2025), Xenium best-practice preprocessing (article).

- Atta L. et al. *Genome Biol* 25 (2024), normalization for targeted panels (article).
- Palla G. et al. *Nat Methods* 19, 171–178 (2022), squidpy.
- Traag V. A. et al. *Sci Rep* 9, 5233 (2019), the Leiden algorithm.